

Interdisciplinary Systems for Site-Specific Spatial Audio: Distributed Sound Architecture and Open Control Protocols

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ABSTRACT

Deploying multichannel spatial audio systems in non-traditional architectural spaces—such as historic industrial warehouses, concrete tunnels, and open outdoor plazas—presents major technical challenges for traditional fixed-channel audio routing. This paper describes a modular, distributed spatial sound architecture built on open network protocols (Open Sound Control, Dante, and WebSockets). We present algorithms for real-time acoustic room correction, dynamic trajectory panning, and collaborative multi-user spatial performance interfaces.

KEY THEOREMS & BOUNDARY EQUATIONS

Formulation [1.1]

Networked Transducer Modularity: Decoupling audio channel count from physical hardware racks through Gigabit Dante networks allows arbitrary scaling of loudspeaker nodes in complex architectural spaces.

Formulation [2.1]

Adaptive Room Correction Invariance: Measuring real-time impulse responses via distributed microphone pods enables dynamic inverse-filtering that neutralizes unwanted architectural echoes.

Formulation [3.1]

Open Protocol Interoperability: Standardizing spatial control messages on high-resolution OSC packets allows seamless integration between Max/MSP, SuperCollider, Unity, and web clients.

1. Escaping the Traditional Concert Hall

Contemporary sound art has largely migrated from sterile concert halls into irregular, site-specific architectural spaces. These environments offer unique acoustic resonance, tactile textures, and spatial character, but break standard 5.1 and 7.1 channel assumptions. Spatial audio must become fluid, reconfigurable, and aware of room boundaries.

2. The Distributed Network Backbone

Our architecture deploys self-powered active loudspeaker pods connected via standard Ethernet switches. Each pod incorporates an internal DSP module, a 24-bit/96 kHz DAC, and a Class-D amplifier. Audio is streamed via Dante with deterministic 0.25 ms latency, while spatial trajectory coordinates are transmitted over OSC.

3. Dynamic Trajectory Panning Software

We developed an open-source spatialization suite running inside Max/MSP and as a standalone C++ audio server. Sound sources can be positioned and animated in 3D cartesian coordinates. The engine automatically calculates speaker weights using Distance-Based Amplitude Panning (DBAP) and High-Order Ambisonics (HOA), gracefully handling asymmetric speaker placements.

4. Public Sound Installations

We document the deployment of a 48-channel sound installation in an abandoned hydroelectric turbine hall. Visitors interacted with the spatial sound field through their mobile phones, shaping localized sonic vortices that drifted across the cavernous steel and concrete architecture.

REFERENCES & PRIMARY CITATIONS

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- Wright, M. & Freed, A. (1997). Open Sound Control: A New Protocol. ICMC.